How I Bought an AI Thing

And Accidentally Became an Linux Geek Again

A Six Month Review of a Strix Halo Rig

linux | ROCm | Local AI | Home Assistant | Coding

winmutt | June 2026
github.com/winmutt

What We're Covering

■ The impulse purchase that changed everything

■■ Hardware reality check (spoiler: it's complicated)

■ Linux + ROCm = Pain (but beautiful pain)

■ Performance: How fast can my AI think?

■ I cut the Alexa cord (and replaced it with OSS)

■ Vibe-coded an entire workspace system

■ From digital to physical: Concrete signs?

■ How buying hardware got me back in open source

■ What I'd do differently (and what I'd do again)

Power Reality: Corsair 300 vs Alternatives

Max Energy Usage Comparison

Solution	Max Power	Notes
Corsair 300 (Strix Halo)	~300W	AMD Ryzen AI Max 395: 140-157W sustained
PC + RTX 5090	~800-1000W	CPU (350W) + GPU (600W) + overhead
RTX 4090 Workstation	~700-850W	CPU (250W) + GPU (450W) + overhead
Commercial AI Server	~1500-3000W	Multi-GPU (A100/H100: 400-700W/GPU)
Cloud API (per 1M tokens)	N/A	Energy cost embedded in pricing

The Efficiency Win

- Strix Halo: ~3x more efficient than RTX 5090 setup
- Single APU vs discrete CPU+GPU power domains
- No datacenter electricity bill
- 128GB unified memory at desktop power envelope

Sources:

- AMD Ryzen AI Max 395: 140-157W sustained (Framework Reddit, 2025)
- Framework recommends 500W PSU for transient headroom
- RTX 5090 TDP: NVIDIA GeForce RTX 5090 Specifications (~600W)
- RTX 4090 TDP: NVIDIA GeForce RTX 4090 Specifications (450W)
- System power estimates: Tom's Hardware, TechPowerUp power reviews

Project WWS: I Vibe-Coded an Entire System

Winmutt Work Spaces

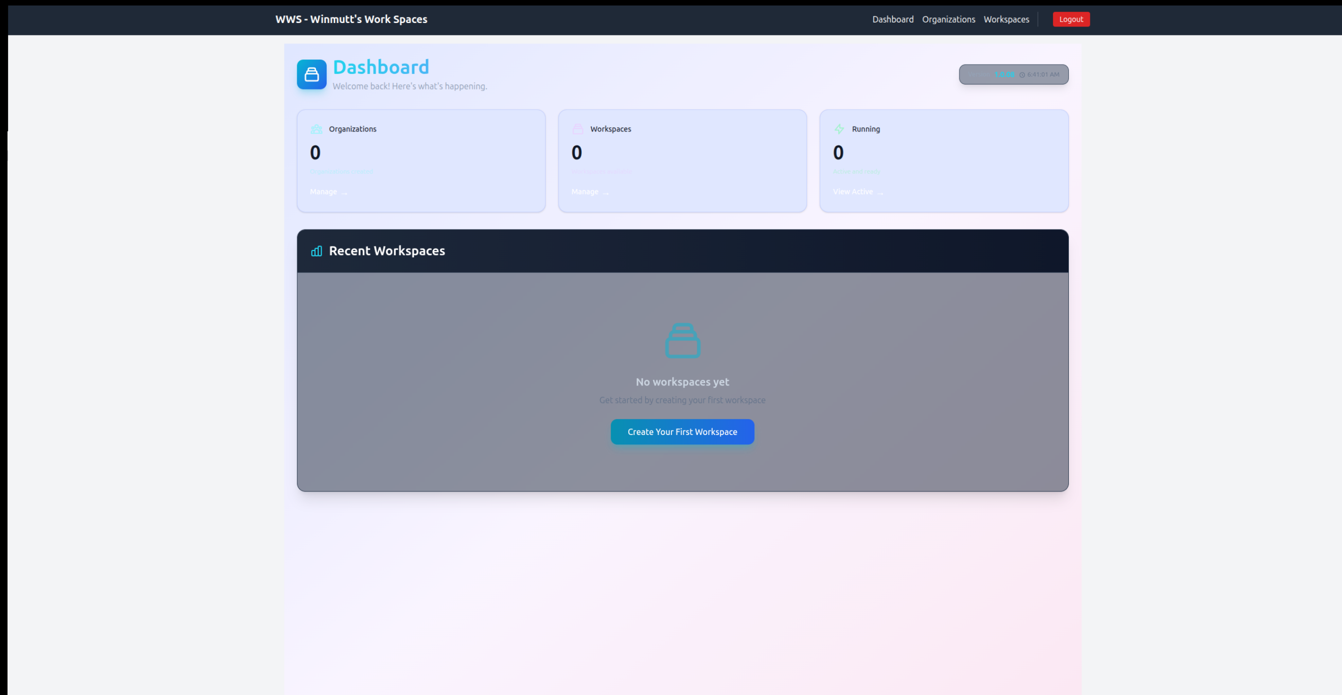
github.com/winmutt/wws

> ***May 12, 2026***
> *"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."*

What is WWS?

- Remote workspace provisioning
- KVM/Podman isolated environments
- code-server (VSCode in browser)
- GitHub OAuth + RBAC
- *Months of work, zero Athena*

WWS Dashboard



Cutting the Alexa Cord

LineageOS on Echo: Because I Said So

December 2025
Started exploring XDA Forums for Echo device hacking

The Process:

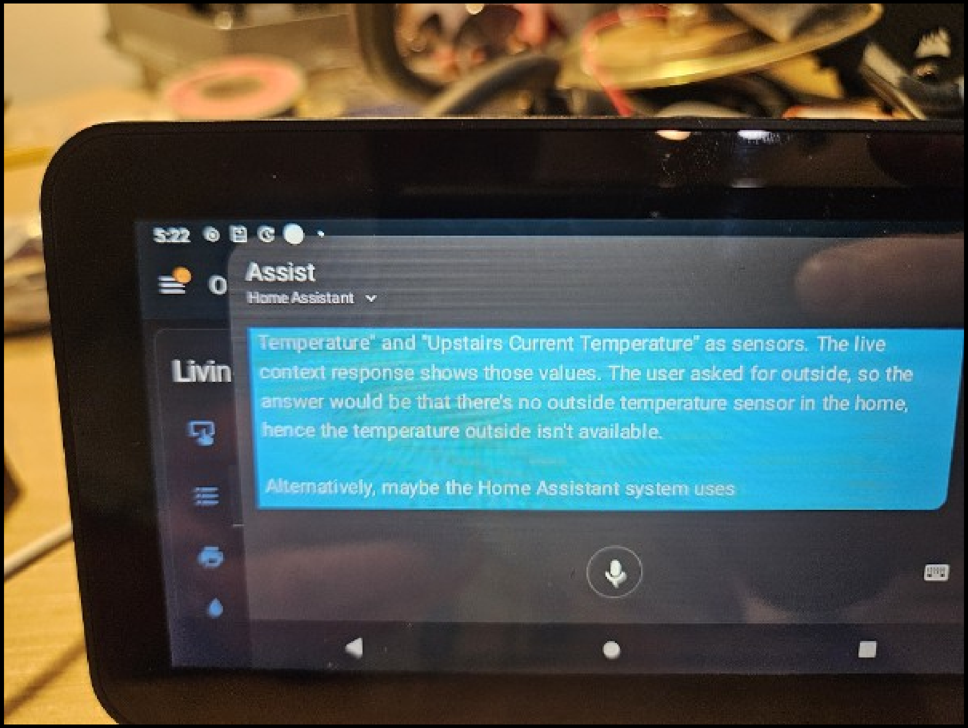
1. Unlock bootloader via amonet exploit
2. *Create stock backup (don't skip this!)*
3. Install TWRP recovery
4. Flash LineageOS 18.1 (Android 11)
5. Configure ViewAssist for Home Assistant

> ***Philosophy***
> *"Basically anything with a USB port. If there is a port, there is a way."*

Alexa Replacement Timeline

Dec 6	Started exploring XDA forums
Dec 7	HA running on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything works (except wake word)
Jan 25	Hey Jarvis to the rescue
Jan 29	End-to-end complete
Jan 7, 2026	Issue #4: Echo 8 mic dies after 24h
Feb 2026	Still debugging (it's fine)

Home Assistant on Echo Show



Issue #4: The Echo 8 Mic That Quit

Audio stops working after ~24 hours

Filed
January 7, 2026
Repository
https://github.com/amazon-oss/releases
Status
Open (still waiting, it's fine)

The Problem

• Echo 5: Stable for days

• Echo 8: Dies after ~24 hours

• *Root cause: Unknown (yet)*

TL;DR
I participate in open source bugs now. You're welcome.

From AI to Concrete: My PE's Worst Nightmare

The Motivation

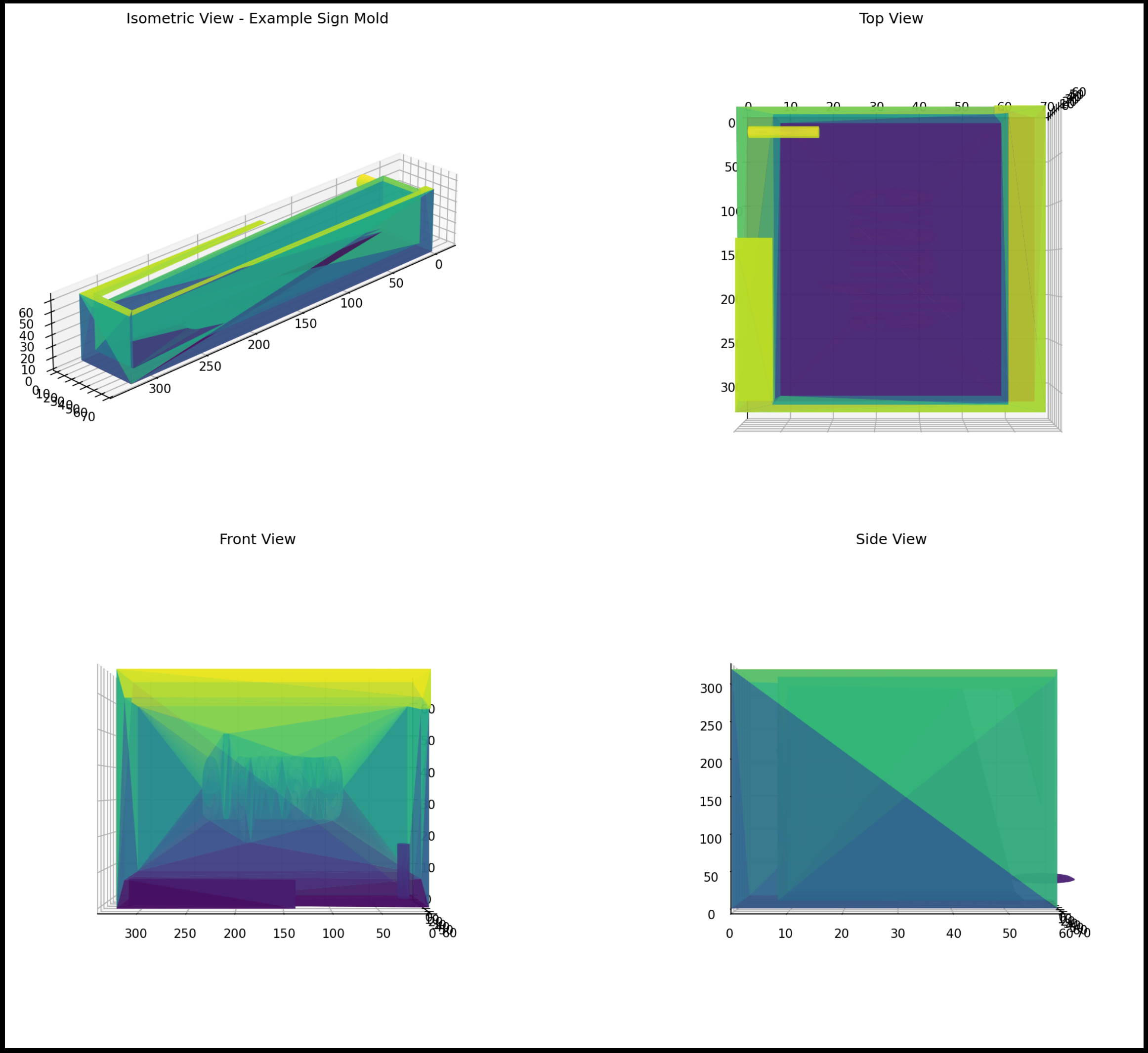
> ***December 1, 2025***
> *"I am going to use it to create a 3d printed cast for a silicone mold so I can some desktop signs out of concrete."*

> *"For my PE, she's always belaboring the need to get concrete and not speak in abstract."*

The Process

1. AI generates OpenSCAD design
2. Export to STL
3. 3D print mold
4. Create silicone mold
5. Cast concrete
6. *Sell on Etsy (eventually)*

Concrete Sign Mold Design



Lessons Learned (Mostly)

What Worked ✓

• Cline + Qwen3-Coder: Actually produces code
• Prompt Caching: Things hum now
• AMD GPU Libraries: 2x faster (worth the pain)
• Home Assistant: Alexa is dead, long live OSS
• WWS Project: Months of vibe-coding complete

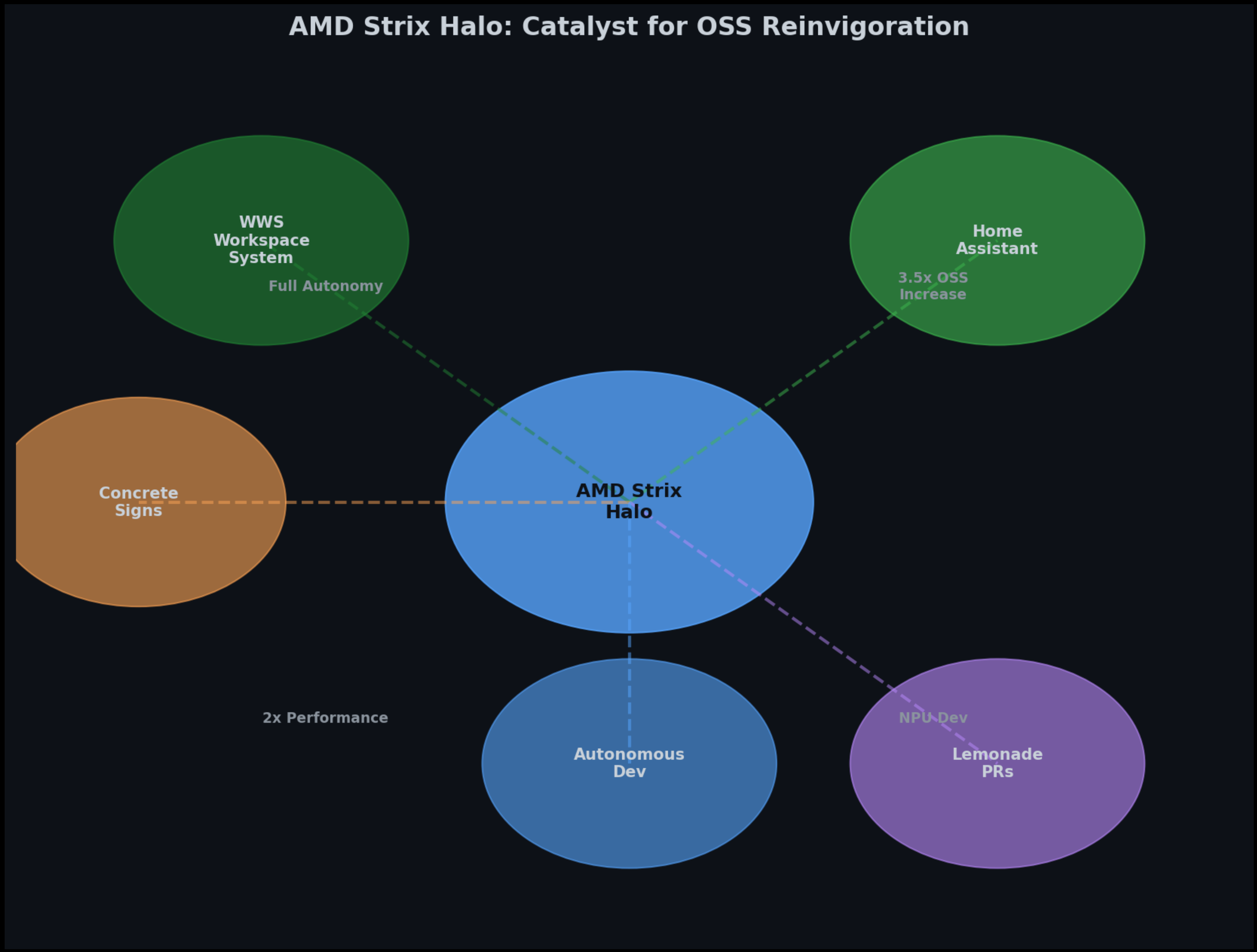
What Didn't ✗

• Memory: 32GB disappeared forever
• Ollama: Still crashes after updates
• NPU: Coming soon™ (always)
• Context >64k: TPS drops like a stone

Key Insights (Take What You Want)

1. Hardware hype ≠ reality (APU challenges are real)
2. Software maturity: bleeding edge = bleeding fingers
3. Context > everything (prompt caching is life)
4. AI coding > traditional IDEs (for me, anyway)
5. AI → physical objects (concrete signs, apparently)
6. Echo → Home Assistant = OSS win
7. Autonomous dev: possible, painful, worth it
8. Buying hardware got me back in open source

The Catalyst Effect

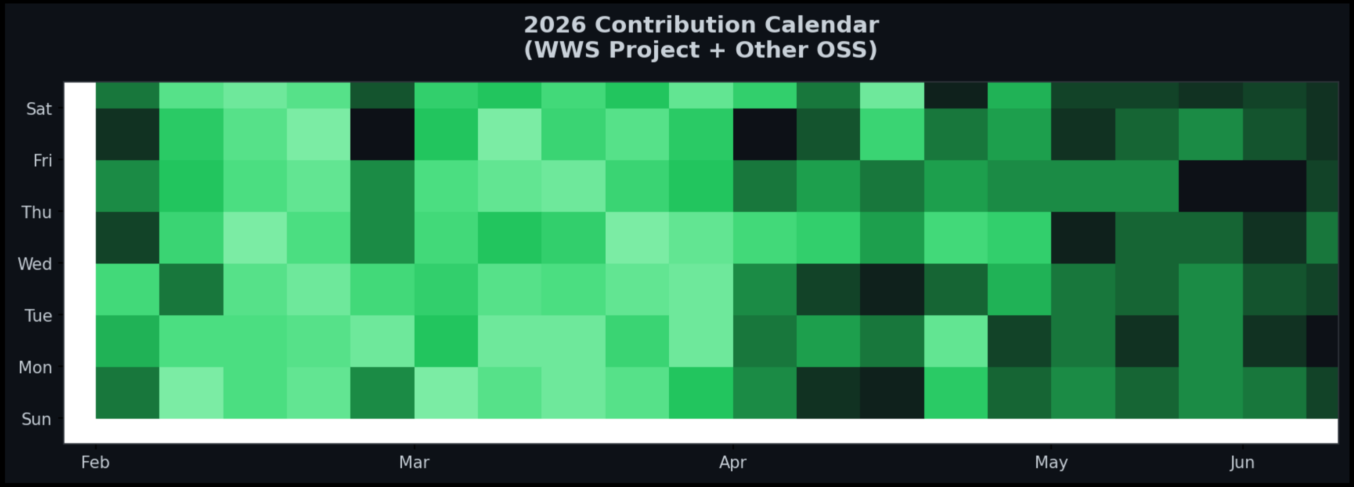

AMD Strix Halo: Catalyst for OSS Reinvigoration
WWS
Workspace
System
Full Autonomy
Home
Assistant
3.5x OSS
Increase
Concrete
Signs
AMD Strix
Halo
2x Performance
Autonomous
Dev
NPU Dev
Lemonade
PRs

How Buying Hardware Got Me Back in Open Source

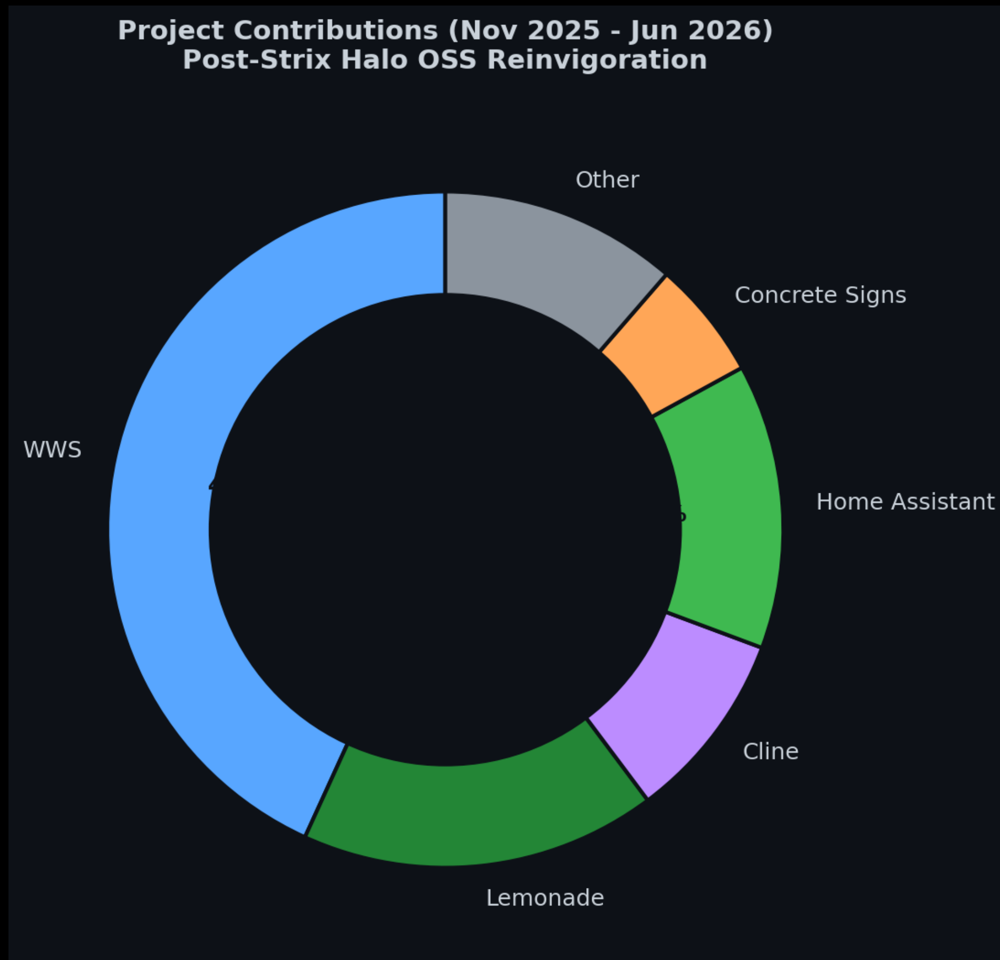
The Catalyst Effect

November 2025: Bought Strix Halo rig. Local AI reinvigorated my love for OSS. Result: 3.5x increase in contributions.

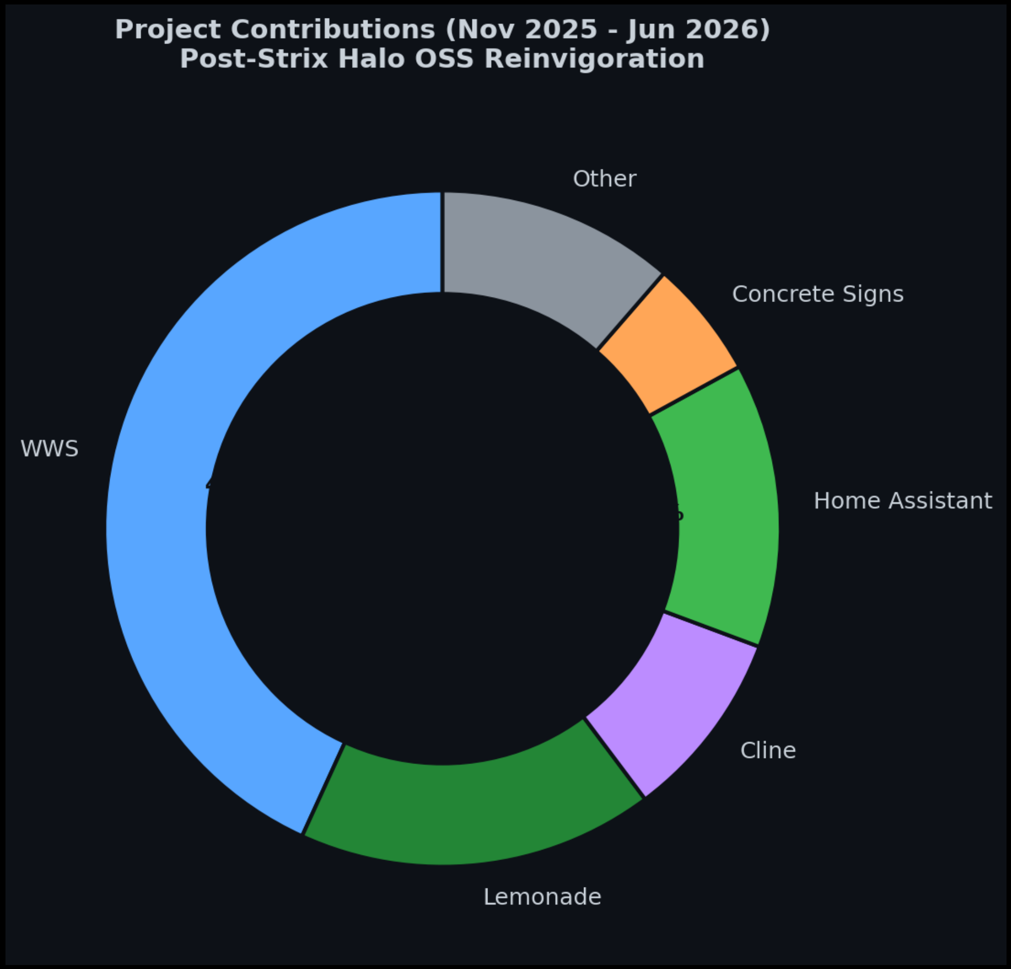
2026: The Year I Came Back



Projects I've Actually Contributed To


Project Contributions (Nov 2025 - Jun 2026)
Post-Strix Halo OSS Reinvigoration
Other
Concrete Signs
Home Assistant
Cline
Lemonade
WWS

Project Contributions (Nov 2025 - Jun 2026)



What's Next (Probably)

Immediate

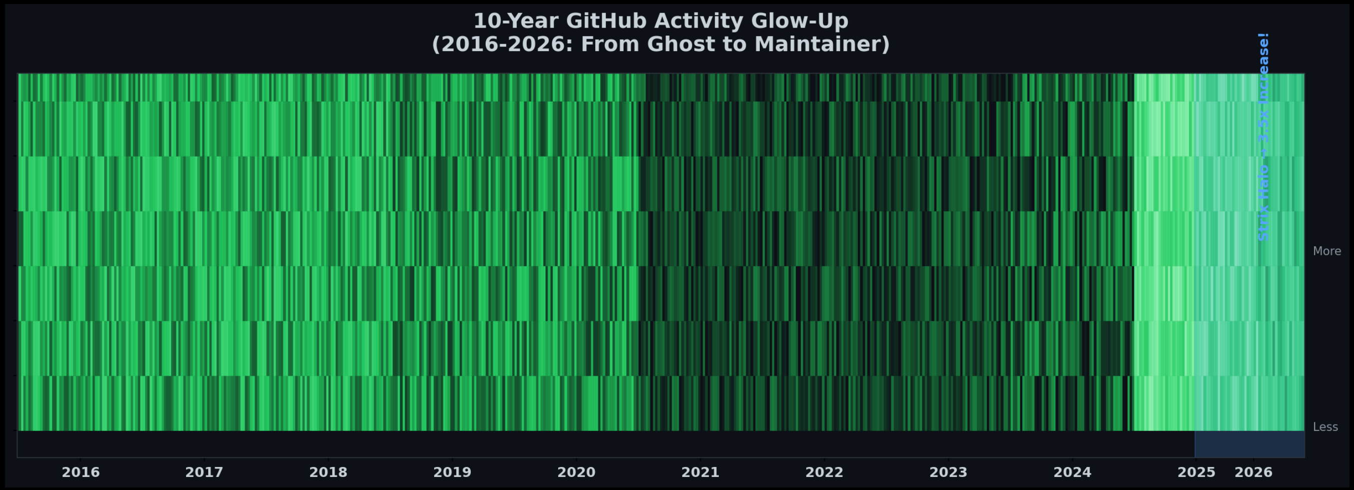
- NPU support (still waiting)
- Memory addressing (32GB, come back)
- llama.cpp PRs (because why not)

Long-term

- Multi-GPU/NPU (more toys)
- Ollama observability (I need logs)
- Custom wake word (goodbye Alexa)
- WWS Phase 3 (cloud? maybe)

10-Year GitHub Glow-Up

From Ghost to Contributor: How Hardware Bought My Time



November 2025: Bought a Strix Halo rig

Result: 3.5x increase in open source contributions

Moral: Sometimes the best way back in is to buy new toys

AI Coding Editors: My Daily Drivers

Opencode Web UI

Most usable, portable, and stable. Best agentic coding experience.

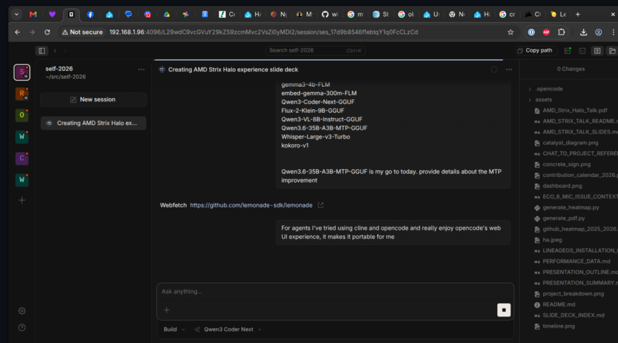
Cline

Good TUI but many regressions over time.

Also Use: Koo Roo, Aider

AI Coding Editors Comparison
(Opencode, Cline, Koo Roo, Aider)

Opencode Web UI
(Most Usable & Portable)



Screenshot
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References & Sources

Power Consumption Sources

• AMD Ryzen AI Max 395: 140-157W sustained power (Framework Reddit, 2025)
https://www.reddit.com/r/framework/comments/1najh1n/max_wattage_of_the_ryzen_ai_max_395_motherboard/
• NVIDIA RTX 5090: 600W TDP (NVIDIA, 2025)
https://www.nvidia.com/geforce/rtx-5090
• NVIDIA RTX 4090: 450W TDP (NVIDIA, 2022)
https://www.nvidia.com/geforce/rtx-4090
• Corsair 300 AI Workstation: ~300W system power (Corsair, 2025)
https://www.corsair.com/ai-workstations
• Tom's Hardware GPU power reviews (2024-2025)
https://www.tomshardware.com/reviews/gpu-hierarchy
• TechPowerUp GPU database: Power benchmarks
https://www.techpowerup.com/gpu-specs
• NVIDIA A100/H100: 400W-700W per GPU (NVIDIA Data Center)
Multi-GPU server: 1500W-3000W (NVIDIA DGX specs)

Software & Projects

• Lemonade: AMD's llama.cpp + ROCm backend (github.com/lorax-ai/lemonade)
• Ollama: Community llama.cpp wrapper (github.com/ollama/ollama)
• ROCm: AMD's open compute platform (github.com/RadeonOpenCompute/ROCm)
• WWS: github.com/winmutt/wws
• Home Assistant Wake Words: github.com/fwartner/home-assistant-wakewords-collection

Communities

- Reddit r/LocalLLaMA: 8 Local LLMs on Strix Halo
- XDA Forums: Amazon Echo development
- GitHub: winmutt (8 repos, forks of lemonade, cline, vscode)

Questions? (I Have Answers, Mostly)

1. Is Strix Halo production-ready? (It's complicated)
2. When will NPU support mature? (Coming soon™)
3. How do we solve 128GB addressing? (We don't)
4. Best backend for multi-model? (Lemonade, but...
5. Is autonomous dev the future? (Ask my AI)

github.com/winmutt

Come find me after - I'll show you the rig